

Calculus II

[6.5.24],

24. Find the approximations T_n , M_n , and S_n to the integral $\int_{-1}^2 xe^x dx$ for $n = 6$ and 12. Then compute the corresponding errors E_T , E_M , and E_S . (Round your answers to six decimal places. You may wish to use the sum command on a computer algebra system.) What observations can you make? In particular, what happens to the errors when n is doubled?

, [7.2.36],

25–37 ■ Find the volume of the described solid S .

36. The base of S is the triangular region with vertices $(0, 0)$, $(3, 0)$, and $(0, 2)$. Cross-sections perpendicular to the y -axis are semicircles.

[7.6.16],

16. Find an equation of the curve that passes through the point $(1, 1)$ and whose slope at (x, y) is y^2/x^3 .

[7.6.40 (a)],

40. Biologists stocked a lake with 400 fish of one species and estimated the species' carrying capacity in the lake to be 10,000. The number of fish tripled in the first year.
- (a) Assuming that the size of the fish population satisfies the logistic equation, find an expression for the size of the population after t years.

[8.5.14],

3–18 ■ Find the radius of convergence and interval of convergence of the series.

14.
$$\sum_{n=1}^{\infty} \frac{(-2)^n}{\sqrt{n}} (x + 3)^n$$

, [8.8.18],

9–16 ■

- (a) Approximate f by a Taylor polynomial with degree n at the number a .
- (b) Use Taylor's Formula to estimate the accuracy of the approximation $f(x) \approx T_n(x)$ when x lies in the given interval.
-  (c) Check your result in part (b) by graphing $|R_n(x)|$.

12. $f(x) = \cos x, \quad a = \pi/3, \quad n = 4, \quad 0 \leq x \leq 2\pi/3$

18. Use the information from Exercise 12 to estimate $\cos 69^\circ$ correct to five decimal places.

9.2.8



8. Find equations of the tangents to the curve $x = \sin t$, $y = \sin(t + \sin t)$ at the origin. Then graph the curve and the tangents.